Question ID: 8adf1335

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A city's total expense budget for one year was x million dollars. The city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between x and y in this context?

Answer

A. $x+y=201$

B. $x-y=201$

C. $2x-y=201$

D. $y-x=201$

Correct Answer: B

Rationale

Choice B is correct. Of the city's total expense budget for one year, the city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. This means that the expression $y+201$ represents the total expense budget, in millions of dollars, for one year. It's given that the total expense budget for one year is x million dollars. It follows then that the expression $y+201$ is equivalent to x, or $y+201=x$. Subtracting y from both sides of this equation yields $201=x-y$. By the symmetric property of equality, this is the same as $x-y=201$.

Choices A and C are incorrect. Because it's given that the total expense budget for one year, x million dollars, is comprised of the departmental expenses, y million dollars, and all other expenses, 201 million dollars, the expressions $x+y$ and $2x-y$ both must be equivalent to a value greater than 201 million dollars. Therefore, the equations $x+y=201$ and $2x-y=201$ aren't true. Choice D is incorrect. The value of x must be greater than the value of y. Therefore, $y-x=201$ can't represent this relationship.